SOC Analyst Project Report SSH Brute Force Attack Detection using Splunk

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1 Introduction

This project demonstrates a real-world Security Operations Center (SOC) workflow using Splunk SIEM. The objective is to simulate, detect, analyze, and respond to an SSH brute force attack.

Architecture:

- Attacker: Kali Linux
- Victim: Ubuntu Server (Splunk Forwarder)
- SIEM: Splunk Enterprise

2 Attack Simulation

A brute force attack was simulated using Hydra from the attacker machine.

2.1 Command Used

```
hydra -l loki31 -P rockyou.txt ssh://192.168.56.20
```

2.2 Objective

To generate multiple failed SSH login attempts and simulate a real attack scenario.

2.3 Screenshot

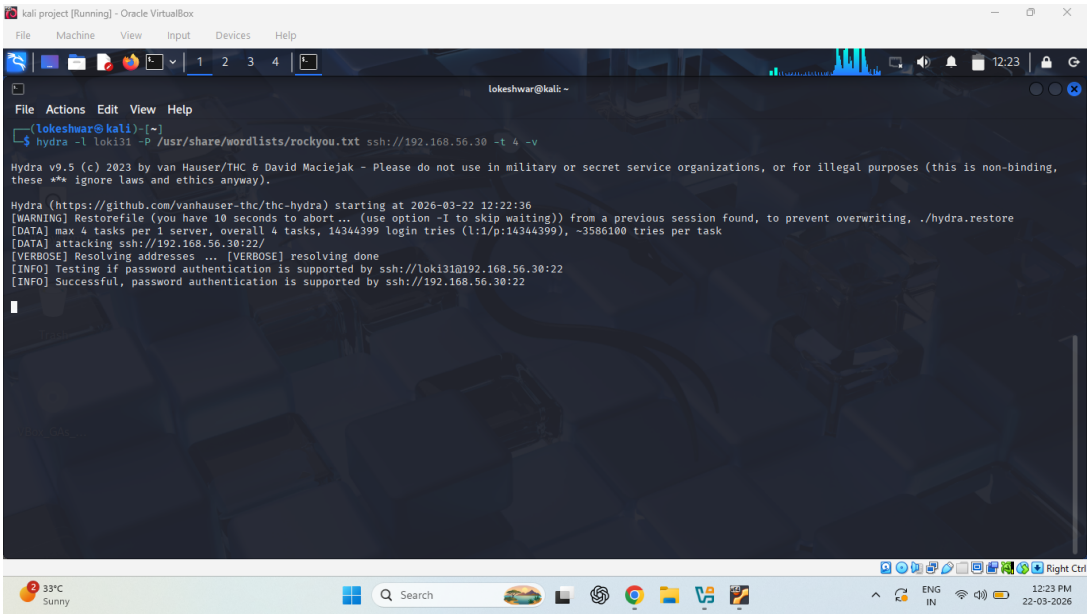

Figure 1: Brute Force Attack using Hydra

3 Detection in Splunk

Logs were ingested into Splunk SIEM using Splunk Universal Forwarder.

3.1 Detection Query

```
index=* source="/var/log/auth.log" "Failed password"
| rex "from (?<src_ip>\d{1,3}(?:.\d{1,3}){3})"
| rex "Failed password for (?:invalid user )?(?<user>[^\s]+)"
| stats count as failed_attempts by src_ip, user
| where failed_attempts >= 20
```

3.2 Result

The query successfully identified the attacker IP and number of failed attempts.

3.3 Screenshot

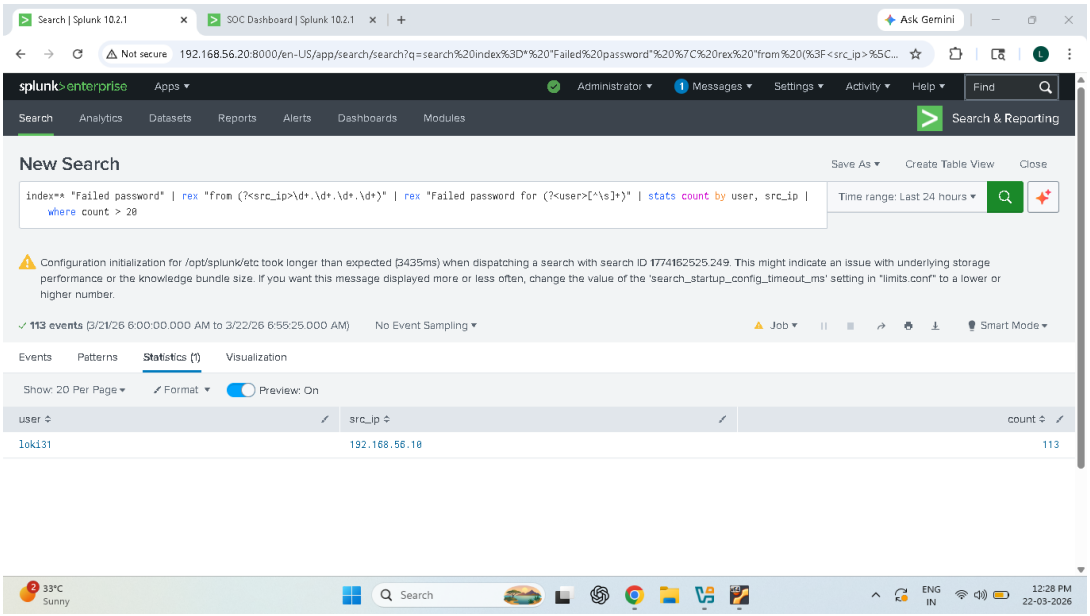

Figure 2: Detection of SSH Brute Force Attack in Splunk

4 Investigation

Further analysis was performed to understand the attack behavior.

4.1 Timeline Analysis

```
index=* source="/var/log/auth.log" src_ip=192.168.56.10
| sort _time
```

4.2 Findings

- Multiple failed login attempts detected
- Possible successful login after brute force
- Attack originated from a single IP

4.3 Screenshot

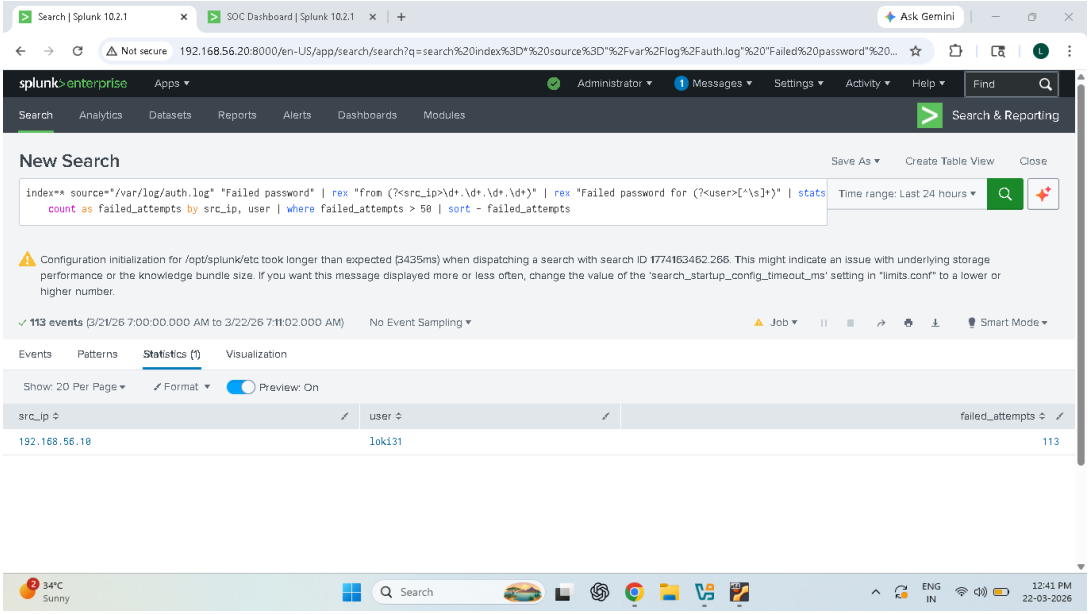

Figure 3: Timeline Analysis of Attack

5 Response Actions

Immediate response actions were taken to mitigate the attack.

5.1 Actions Performed

```
sudo ufw deny from 192.168.56.10
sudo passwd -l loki31
```

5.2 Outcome

- Attacker IP blocked
- Compromised account secured

5.3 Screenshot

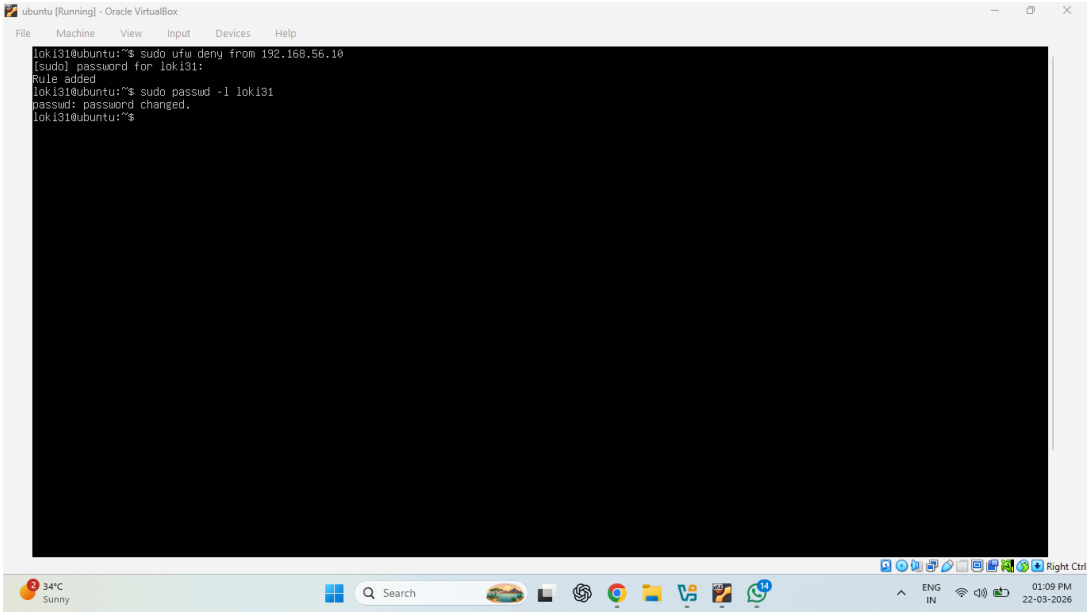

Figure 4: Response Actions on Victim Machine

6 Dashboard Visualization

A SOC dashboard was created in Splunk to monitor attack patterns.

6.1 Panels

- Failed login attempts over time
- Top attacker IPs
- Successful vs Failed logins

6.2 Screenshot

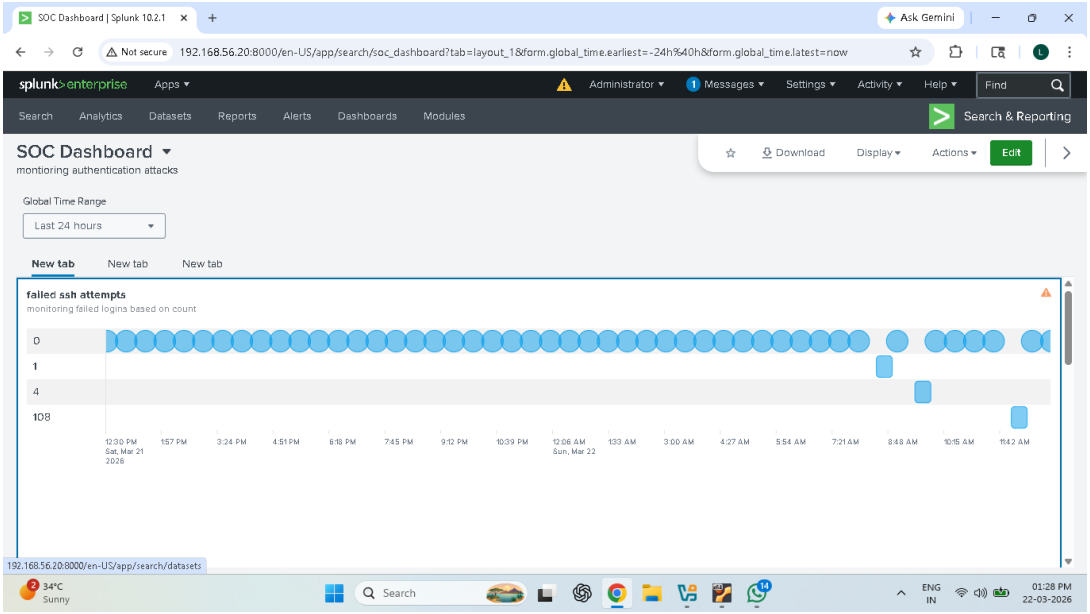

Figure 5: SOC Dashboard in Splunk

6.3 Screenshot

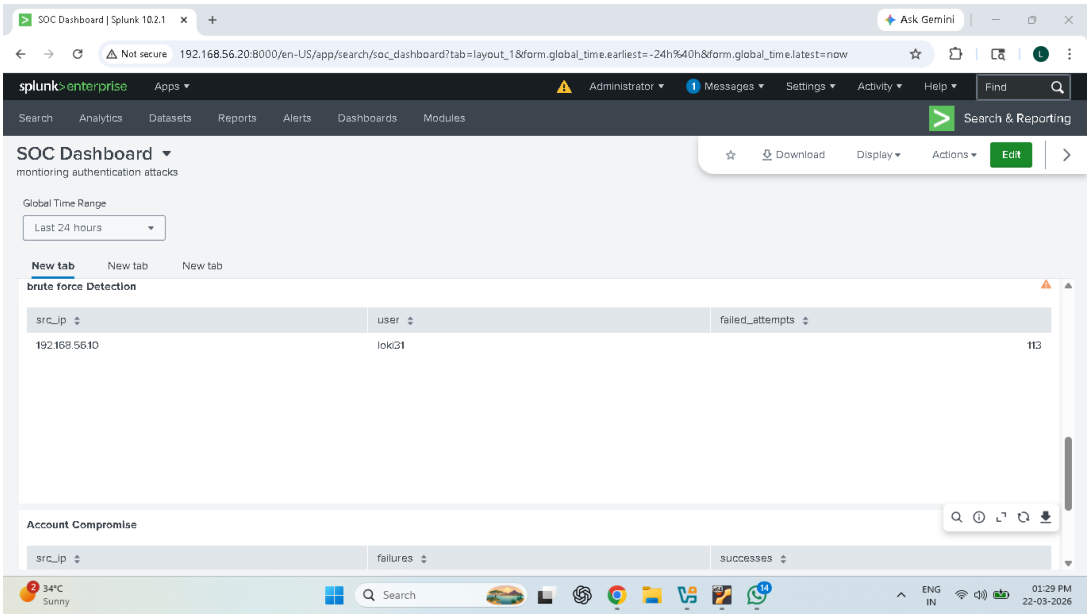

Figure 6: SOC Dashboard in Splunk

6.4 Screenshot

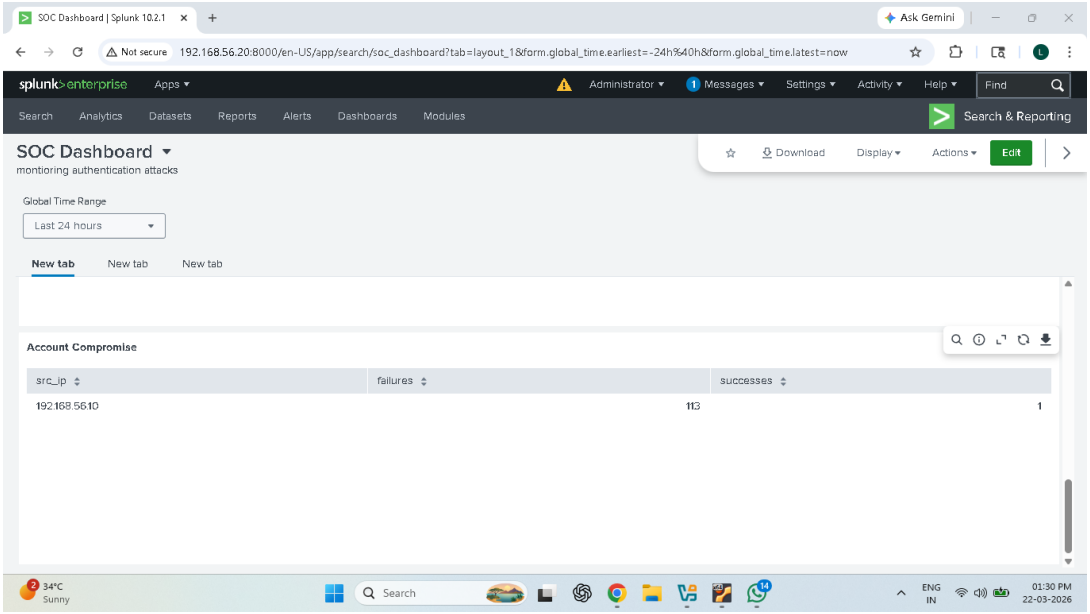

Figure 7: SOC Dashboard in Splunk

7 Conclusion

This project successfully demonstrated the complete SOC workflow:

- Attack Simulation
- Detection using Splunk
- Log Analysis and Investigation
- Incident Response
- Reporting

This hands-on experience reflects real-world SOC analyst responsibilities and strengthens skills in SIEM-based threat detection.